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Isolated and Exhausted: Attacking Operating Systems via Site Isolation in the Browser

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Site Isolation is a security architecture for browsers to protect against side-channel and renderer exploits by separating content from different sites at the operating system (OS) process level. By aligning web and OS security boundaries, Site Isolation promises to defend against these attack classes systematically in a streamlined architecture. However, Site Isolation is a large-scale architectural change that also makes OS resources more accessible to web attackers, and thus exposes web users to new risks at the OS level. In this paper, we present the first systematic study of OS resource exhaustion attacks based on Site Isolation, in the web attacker model, in three steps: (1) first-level resources directly accessible with Site Isolation; (2) second-level resources whose direct use is protected by the browser sandbox; (3) an advanced, real-world attack. For (1) we show how to create a fork bomb, highlighting conceptual gaps in the Site Isolation architecture. For (2) we show how to block all UDP sockets in an OS, using a variety of advanced browser features. For (3), we implement a fully working DNS Cache Poisoning attack based on Site Isolation, building on (2) and bypassing a major security feature of DNS. Our results show that the interplay between modern browser features and older OS features is increasingly problematic and needs further research.

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